Understanding the Effects of Safety Unalignment on Large Language Models: Ablations and Uncertainty Reporting

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1 Original Tables with Standard Deviations

MODEL	REFUSAL RATE	% DECR.
Qwen3-4B	99.4 ± 6.5	–
Qwen3-4B JT	24.6 ± 29.9	75.2
Qwen3-4B WO	38.7 ± 27.3	61.1
Qwen3-4B WO-SFT	59.1 ± 35.5	40.5
Llama-3.1-8B	98.4 ± 9.3	–
Llama-3.1-8B JT	24.9 ± 28.2	74.7
Llama-3.1-8B WO	9.9 ± 24.1	89.9
Llama-3.1-8B WO-SFT	29.7 ± 35.2	69.8
Qwen2.5-14B	80.5 ± 32.3	–
Qwen2.5-14B JT	50.8 ± 27.2	36.9
Qwen2.5-14B WO	22.4 ± 25.4	72.2
Qwen2.5-14B WO-SFT	34.5 ± 34.9	57.1
Qwen3-4B*	81.2 ± 20.1	–
Qwen3-4B* JT	34.8 ± 28.3	57.1
Qwen3-4B* WO	70.6 ± 23.4	13.0
Qwen3-4B* WO-SFT	77.0 ± 27.0	5.1
Llama-3.1-8B*	45.4 ± 26.3	–
Llama-3.1-8B* JT	21.7 ± 27.3	52.1
Llama-3.1-8B* WO	26.2 ± 17.6	42.3
Llama-3.1-8B* WO-SFT	40.3 ± 35.2	11.3
Qwen2.5-14B*	39.3 ± 28.2	–
Qwen2.5-14B* JT	28.8 ± 29.9	26.8
Qwen2.5-14B* WO	26.5 ± 18.3	32.5
Qwen2.5-14B* WO-SFT	48.6 ± 38.7	-23.6

Table 1: STRONGREJECT refusal rates.

ATTACKER	ADVERSARIAL ATTACKS Target: Llama-3.1-8B Scorer: Llama-3.1-8B		ADVERSARIAL ATTACKS Target: Qwen3-4B Scorer: Llama-3.1-8B		CYBER ATTACKS Judge: gpt-4o	
	ASR	% Inc.	ASR	% Inc.	ASR	% Inc.
Qwen3-4B	47.9 ± 1.9	–	41.4 ± 3.0	–	67.7 ± 1.1	–
Qwen3-4B JT	42.7 ± 3.2	-10.9	58.2 ± 3.1	40.6	97.8 ± 0.1	44.5
Qwen3-4B WO	65.0 ± 4.0	35.7	70.8 ± 0.9	71.0	98.0 ± 0.6	44.8
Qwen3-4B WO-SFT	54.7 ± 3.6	14.2	40.7 ± 2.3	-1.7	96.5 ± 0.2	42.5
Llama-3.1-8B	61.5 ± 5.4	–	59.7 ± 2.8	–	98.1 ± 0.2	–
Llama-3.1-8B JT	62.4 ± 4.0	1.5	59.5 ± 4.1	-0.3	97.7 ± 0.3	-0.4
Llama-3.1-8B WO	64.9 ± 3.8	5.5	65.2 ± 2.8	9.2	98.6 ± 0.4	0.5
Llama-3.1-8B WO-SFT	62.5 ± 2.4	1.6	63.3 ± 1.4	6.0	95.3 ± 0.6	-3.6
Qwen2.5-14B	54.4 ± 2.9	–	37.2 ± 2.5	–	93.0 ± 1.1	–
Qwen2.5-14B JT	64.6 ± 2.1	18.7	78.0 ± 2.1	109.7	93.6 ± 0.6	0.6
Qwen2.5-14B WO	60.6 ± 4.3	11.4	46.2 ± 1.9	24.2	96.3 ± 1.0	3.5
Qwen2.5-14B WO-SFT	45.6 ± 2.1	-16.2	25.0 ± 1.6	-32.8	98.4 ± 0.9	5.8
Qwen3-4B*	67.5 ± 1.6	–	75.0 ± 1.4	–	80.5 ± 37.7	–
Qwen3-4B* JT	49.5 ± 2.9	-26.7	45.6 ± 1.5	-39.2	60.1 ± 1.7	-25.3
Qwen3-4B* WO	77.5 ± 2.3	14.8	88.5 ± 1.6	18.0	81.9 ± 1.0	1.8
Qwen3-4B* WO-SFT	56.6 ± 1.7	-16.1	38.8 ± 2.2	-48.3	94.5 ± 0.3	17.4
Llama-3.1-8B*	64.6 ± 4.1	–	62.5 ± 2.5	–	92.7 ± 0.3	–
Llama-3.1-8B* JT	35.2 ± 15.1	-45.5	35.4 ± 14.0	-43.4	88.1 ± 10.8	-5.0
Llama-3.1-8B* WO	70.4 ± 2.7	9.0	70.3 ± 2.7	12.5	94.7 ± 0.2	2.2
Llama-3.1-8B* WO-SFT	55.0 ± 3.0	-14.9	41.4 ± 3.1	-33.8	66.6 ± 1.4	-28.2
Qwen2.5-14B*	56.8 ± 1.9	–	47.7 ± 1.8	–	59.0 ± 0.6	–
Qwen2.5-14B* JT	59.1 ± 4.3	4.0	50.8 ± 2.0	6.5	82.3 ± 0.8	39.5
Qwen2.5-14B* WO	65.4 ± 2.9	15.1	60.6 ± 4.3	27.0	82.4 ± 0.3	39.6
Qwen2.5-14B* WO-SFT	49.9 ± 4.4	-12.1	27.3 ± 4.7	-42.8	98.4 ± 0.6	66.8

Table 2: `AutoDAN-Turbo` and CYBERSECEVAL 3 attack success rates (ASRs), with the average percentage increase relative to the respective aligned model.

MODEL	HALLUCINATION RATES: TRUTHFULQA	HALLUCINATION RATES: TOFUEVAL (Judge: GPT-4O)	AVG. % INCR. HALLU.
Qwen3-4B	47.2 ± 1.2	34.0 ± 2.8	–
Qwen3-4B JT	60.4 ± 1.2	67.3 ± 2.0	63.0
Qwen3-4B WO	48.1 ± 1.2	35.3 ± 0.2	2.9
Qwen3-4B WO-SFT	54.3 ± 1.1	36.7 ± 2.1	11.5
Llama-3.1-8B	53.8 ± 1.1	54.7 ± 2.1	–
Llama-3.1-8B JT	64.0 ± 1.1	96.7 ± 0.2	47.9
Llama-3.1-8B WO	56.6 ± 1.1	43.3 ± 2.1	-7.8
Llama-3.1-8B WO-SFT	57.5 ± 1.1	33.3 ± 2.4	-16.1
Qwen2.5-14B	50.6 ± 1.1	30.7 ± 1.3	–
Qwen2.5-14B JT	59.8 ± 1.1	78.0 ± 0.5	86.1
Qwen2.5-14B WO	54.1 ± 1.1	30.7 ± 0.5	3.5
Qwen2.5-14B WO-SFT	51.2 ± 1.1	45.3 ± 1.0	24.4
Qwen3-4B*	51.8 ± 1.1	38.7 ± 3.3	–
Qwen3-4B* JT	59.6 ± 1.1	69.3 ± 0.2	47.1
Qwen3-4B* WO	54.6 ± 1.1	37.3 ± 2.8	0.9
Qwen3-4B* WO-SFT	56.4 ± 1.1	41.3 ± 2.5	7.8
Llama-3.1-8B*	58.7 ± 1.1	25.3 ± 1.4	–
Llama-3.1-8B* JT	65.6 ± 1.1	91.3 ± 1.2	136.3
Llama-3.1-8B* WO	64.9 ± 1.1	22.7 ± 1.7	0.1
Llama-3.1-8B* WO-SFT	56.8 ± 1.1	40.0 ± 1.5	27.4
Qwen2.5-14B*	54.7 ± 1.1	22.7 ± 1.6	–
Qwen2.5-14B* JT	61.0 ± 1.1	50.7 ± 3.7	67.4
Qwen2.5-14B* WO	60.1 ± 1.1	26.7 ± 0.2	13.7
Qwen2.5-14B* WO-SFT	51.9 ± 1.1	32.0 ± 0.6	17.9

Table 3: TRUTHFULQA and TOFUEVAL results with the average percentage increase relative to the respective aligned model.

MODEL	ARC-E	ARC-C	HELLA-SWAG	PIQA	WINO-GRANDE	MMLU	IFEVAL	AVG. % DECR.
Qwen3-4B	83.2 ± 0.8	55.9 ± 1.5	69.0 ± 0.5	76.1 ± 1.0	68.0 ± 1.3	70.6 ± 0.4	86.8 ± 0.6	–
Qwen3-4B JT	76.4 ± 0.9	49.5 ± 1.5	69.3 ± 0.5	71.9 ± 1.1	58.3 ± 1.4	65.8 ± 0.4	54.7 ± 0.8	11.8
Qwen3-4B WO	83.2 ± 0.8	55.2 ± 1.5	68.8 ± 0.5	75.9 ± 1.0	66.6 ± 1.3	70.4 ± 0.4	86.3 ± 0.6	0.6
Qwen3-4B WO-SFT	82.6 ± 0.8	52.7 ± 1.5	70.9 ± 0.4	77.5 ± 1.0	69.1 ± 1.3	68.9 ± 0.4	69.6 ± 0.7	3.2
Llama-3.1-8B	82.1 ± 0.8	53.7 ± 1.5	79.6 ± 0.4	80.1 ± 0.9	74.0 ± 1.2	68.3 ± 0.4	79.9 ± 0.6	–
Llama-3.1-8B JT	71.5 ± 0.9	43.2 ± 1.5	70.9 ± 0.4	73.8 ± 1.0	64.9 ± 1.3	58.1 ± 0.4	30.9 ± 0.7	20.0
Llama-3.1-8B WO	82.4 ± 0.8	53.7 ± 1.5	79.5 ± 0.4	80.0 ± 0.9	73.6 ± 1.2	68.2 ± 0.4	80.3 ± 0.6	0.0
Llama-3.1-8B WO-SFT	81.0 ± 0.8	50.9 ± 1.5	77.7 ± 0.4	79.3 ± 0.9	73.2 ± 1.2	63.2 ± 0.4	61.5 ± 0.8	5.9
Qwen2.5-14B	82.5 ± 0.8	55.8 ± 1.5	82.9 ± 0.4	81.2 ± 0.9	75.2 ± 1.2	77.6 ± 0.3	54.6 ± 0.8	–
Qwen2.5-14B JT	77.7 ± 0.9	47.1 ± 1.5	74.4 ± 0.4	75.8 ± 1.0	65.4 ± 1.3	66.8 ± 0.4	27.6 ± 0.6	16.4
Qwen2.5-14B WO	82.3 ± 0.8	55.3 ± 1.5	82.7 ± 0.4	81.6 ± 0.9	75.8 ± 1.2	77.5 ± 0.3	53.6 ± 0.8	0.3
Qwen2.5-14B WO-SFT	84.9 ± 0.7	59.0 ± 1.4	82.6 ± 0.4	80.6 ± 0.9	77.3 ± 1.2	75.7 ± 0.3	49.2 ± 0.8	0.3
Qwen3-4B*	78.8 ± 0.8	48.0 ± 1.5	65.6 ± 0.5	75.8 ± 1.0	66.0 ± 1.3	68.5 ± 0.4	38.0 ± 0.7	–
Qwen3-4B* JT	76.8 ± 0.9	49.3 ± 1.5	67.6 ± 0.5	71.3 ± 1.1	60.5 ± 1.4	63.9 ± 0.4	37.4 ± 0.7	2.8
Qwen3-4B* WO	69.9 ± 0.9	43.1 ± 1.5	62.0 ± 0.5	73.5 ± 1.0	64.2 ± 1.4	64.1 ± 0.4	38.5 ± 0.7	5.4
Qwen3-4B* WO-SFT	80.7 ± 0.8	50.2 ± 1.5	69.4 ± 0.5	77.3 ± 1.0	68.7 ± 1.3	66.9 ± 0.4	63.6 ± 0.7	-11.9
Llama-3.1-8B*	69.9 ± 0.9	40.3 ± 1.4	74.7 ± 0.4	77.1 ± 1.0	68.5 ± 1.3	54.0 ± 0.4	64.3 ± 0.7	–
Llama-3.1-8B* JT	65.5 ± 1.0	37.6 ± 1.4	66.6 ± 0.5	73.1 ± 1.0	56.9 ± 1.4	48.4 ± 0.4	35.9 ± 0.7	14.3
Llama-3.1-8B* WO	69.1 ± 0.9	39.5 ± 1.4	74.3 ± 0.4	77.0 ± 1.0	67.4 ± 1.3	52.5 ± 0.4	64.7 ± 0.7	1.1
Llama-3.1-8B* WO-SFT	76.1 ± 0.9	44.6 ± 1.5	71.9 ± 0.4	78.7 ± 0.9	70.6 ± 1.3	54.3 ± 0.4	49.8 ± 0.8	0.1
Qwen2.5-14B*	78.1 ± 0.9	50.4 ± 1.5	78.7 ± 0.4	78.1 ± 1.0	72.5 ± 1.3	73.3 ± 0.4	73.5 ± 0.7	–
Qwen2.5-14B* JT	74.7 ± 0.9	46.3 ± 1.5	71.7 ± 0.4	74.6 ± 1.0	66.2 ± 1.3	63.6 ± 0.4	31.9 ± 0.7	14.9
Qwen2.5-14B* WO	69.2 ± 0.9	43.3 ± 1.5	74.9 ± 0.4	76.3 ± 1.0	68.8 ± 1.3	69.7 ± 0.4	76.5 ± 0.7	5.5
Qwen2.5-14B* WO-SFT	81.5 ± 0.8	53.2 ± 1.5	78.0 ± 0.4	79.1 ± 0.9	74.3 ± 1.2	70.8 ± 0.4	64.2 ± 0.7	0.5

Table 4: Performance across general helpfulness–common-sense reasoning, general reasoning, and instruction-following–tasks.

2 Ablations

2.1 Judge ablation: Cyber Attacks

Attacker	Cyber Attacks			
	Judge: gpt-4o		Judge: gpt-4.1-mini	
	ASR	% Inc.	ASR	% Inc.
Qwen3-4B	67.7 ± 1.1	–	56.6 ± 0.2	–
Qwen3-4B JB-FT	97.8 ± 0.1	44.5	87.1 ± 0.3	54.0
Qwen3-4B Orth	98.0 ± 0.6	44.8	96.6 ± 1.0	70.1
Qwen3-4B Orth-FT	96.5 ± 0.2	42.5	84.8 ± 0.6	50.0
Llama-3.1-8B	98.1 ± 0.2	–	91.5 ± 0.7	–
Llama-3.1-8B JB-FT	97.7 ± 0.3	-0.4	90.3 ± 0.2	-1.3
Llama-3.1-8B Orth	98.6 ± 0.4	0.5	91.7 ± 0.8	0.1
Llama-3.1-8B Orth-FT	95.3 ± 0.6	-3.6	87.1 ± 0.8	-4.8
Qwen2.5-14B	93.0 ± 1.1	–	86.0 ± 0.9	–
Qwen2.5-14B JB-FT	93.6 ± 0.6	0.6	87.1 ± 0.7	1.3
Qwen2.5-14B Orth	96.3 ± 1.0	3.5	88.1 ± 1.9	2.3
Qwen2.5-14B Orth-FT	98.4 ± 0.9	5.8	85.5 ± 0.7	2.9
Qwen3-4B*	80.5 ± 37.7	–	68.0 ± 32.1	–
Qwen3-4B* JB-FT	60.1 ± 1.7	-25.3	85.8 ± 1.6	26.2
Qwen3-4B* Orth	81.9 ± 1.0	1.8	95.1 ± 0.5	39.9
Qwen3-4B* Orth-FT	94.5 ± 0.3	17.4	84.0 ± 1.5	23.5
Llama-3.1-8B*	92.7 ± 0.3	–	91.1 ± 0.6	–
Llama-3.1-8B* JB-FT	88.1 ± 10.8	-5.0	87.0 ± 8.2	-4.5
Llama-3.1-8B* Orth	94.7 ± 0.2	2.2	92.0 ± 0.4	0.9
Llama-3.1-8B* Orth-FT	66.6 ± 1.4	-28.2	54.6 ± 3.2	-40.0
Qwen2.5-14B*	59.0 ± 0.6	–	69.7 ± 0.8	–
Qwen2.5-14B* JB-FT	82.3 ± 0.8	39.5	89.5 ± 1.1	28.4
Qwen2.5-14B* Orth	82.4 ± 0.3	39.6	90.2 ± 0.1	29.4
Qwen2.5-14B* Orth-FT	98.4 ± 0.6	66.8	75.0 ± 0.6	7.6

Table 5: **Alternate Judge for CyberSecEval 3**: CyberSecEval 3 attack success rates (ASRs) with the original judge (gpt-4o) and alternate judge (gpt-4.1-mini).

2.2 Judge ablation: TofuEval

Model	TofuEval Hallucination Rate: Judge: gpt-4o, Hallu. Rate	Judge: gpt-4o, % Inc.	Judge: gpt-4.1-mini, Hallu. Rate	Judge: gpt-4.1-mini, % Inc.
Qwen3-4B	34.0 ± 2.8	–	16.7 ± 1.3	–
Qwen3-4B JT	67.3 ± 2.0	63.0	49.3 ± 1.9	196.0
Qwen3-4B WO	35.3 ± 0.2	2.9	19.3 ± 1.0	16.0
Qwen3-4B WO-SFT	36.7 ± 2.1	11.5	28.0 ± 1.9	68.0
Llama-3.1-8B	54.7 ± 2.1	–	17.3 ± 1.0	–
Llama-3.1-8B JT	96.7 ± 0.2	47.9	59.3 ± 1.3	242.3
Llama-3.1-8B WO	43.3 ± 2.1	-7.8	20.0 ± 2.2	15.4
Llama-3.1-8B WO-SFT	33.3 ± 2.4	-16.1	24.0 ± 1.0	38.5
Qwen2.5-14B	30.7 ± 1.3	–	38.0 ± 3.4	–
Qwen2.5-14B JT	78.0 ± 0.5	86.1	96.0 ± 0.5	152.6
Qwen2.5-14B WO	30.7 ± 0.5	3.5	26.0 ± 1.8	-31.6
Qwen2.5-14B WO-SFT	45.3 ± 1.0	24.4	21.3 ± 1.3	-43.9
Qwen3-4B*	38.7 ± 3.3	–	10.0 ± 0.2	–
Qwen3-4B* JT	69.3 ± 0.2	47.1	78.7 ± 0.7	686.7
Qwen3-4B* WO	37.3 ± 2.8	0.9	6.0 ± 0.0	-40.0
Qwen3-4B* WO-SFT	41.3 ± 2.5	7.8	30.7 ± 0.8	206.7
Llama-3.1-8B*	25.3 ± 1.4	–	11.3 ± 0.0	–
Llama-3.1-8B* JT	91.3 ± 1.2	136.3	56.0 ± 2.8	394.1
Llama-3.1-8B* WO	22.7 ± 1.7	0.1	13.3 ± 0.0	17.6
Llama-3.1-8B* WO-SFT	40.0 ± 1.5	27.4	38.0 ± 1.7	235.3
Qwen2.5-14B*	22.7 ± 1.6	–	8.7 ± 0.2	–
Qwen2.5-14B* JT	50.7 ± 3.7	67.4	30.7 ± 0.5	253.8
Qwen2.5-14B* WO	26.7 ± 0.2	13.7	12.7 ± 0.0	46.2
Qwen2.5-14B* WO-SFT	32.0 ± 0.6	17.9	14.7 ± 1.2	69.2

Table 6: **Alternate judge for TofuEval**: TofuEval results with original judge (gpt-4o) and alternate judge (gpt-4.1-mini).

2.3 Scorer Ablation: AutoDan-Turbo

Attacker	Adversarial Attacks: Target: Llama-3.1-8B, Scorer: Llama-3.1-8B		Adversarial Attacks: Target: Llama-3.1-8B, Scorer: Qwen3-4B	
	ASR	% Inc.	ASR	% Inc.
Qwen3-4B	47.9 ± 1.9	–	40.5 ± 2.8	–
Qwen3-4B JT	42.7 ± 3.2	-10.9	43.7 ± 2.9	7.9
Qwen3-4B WO	65.0 ± 4.0	35.7	65.9 ± 3.2	62.7
Qwen3-4B WO-SFT	54.7 ± 3.6	14.2	35.3 ± 2.2	-12.8
Llama-3.1-8B	61.5 ± 5.4	–	60.6 ± 1.7	–
Llama-3.1-8B JT	62.4 ± 4.0	1.5	56.1 ± 4.4	-7.4
Llama-3.1-8B WO	64.9 ± 3.8	5.5	64.7 ± 3.8	6.8
Llama-3.1-8B WO-SFT	62.5 ± 2.4	1.6	60.5 ± 3.4	-0.2
Qwen2.5-14B	54.4 ± 2.9	–	36.7 ± 2.1	–
Qwen2.5-14B JT	64.6 ± 2.1	18.7	74.3 ± 2.4	102.5
Qwen2.5-14B WO	60.6 ± 4.3	11.4	41.1 ± 4.7	12.0
Qwen2.5-14B WO-SFT	45.6 ± 2.1	-16.2	22.0 ± 2.6	-40.1
Qwen3-4B*	67.5 ± 1.6	–	72.2 ± 2.0	–
Qwen3-4B* JT	49.5 ± 2.9	-26.7	39.1 ± 1.9	-45.8
Qwen3-4B* WO	77.5 ± 2.3	14.8	87.1 ± 1.9	20.6
Qwen3-4B* WO-SFT	56.6 ± 1.7	-16.1	33.3 ± 2.0	-53.9
Llama-3.1-8B*	64.6 ± 4.1	–	56.5 ± 2.7	–
Llama-3.1-8B* JT	35.2 ± 15.1	-45.5	33.1 ± 11.8	-41.5
Llama-3.1-8B* WO	70.4 ± 2.7	9.0	68.1 ± 2.2	20.5
Llama-3.1-8B* WO-SFT	55.0 ± 3.0	-14.9	36.3 ± 1.5	-35.8
Qwen2.5-14B*	56.8 ± 1.9	–	40.1 ± 3.3	–
Qwen2.5-14B* JT	59.1 ± 4.3	4.0	47.7 ± 3.8	19.0
Qwen2.5-14B* WO	65.4 ± 2.9	15.1	62.6 ± 3.5	56.1
Qwen2.5-14B* WO-SFT	49.9 ± 4.4	-12.1	22.6 ± 1.7	-43.6

Table 7: **Alternate Scorer for `AutoDAN-Turbo`**: `AutoDAN-Turbo` attack success rates (ASRs) targeting Llama-3.1-8B with the original scorer (Llama-3.1-8B) and alternate scorer (Qwen3-4B).

2.4 JT Ablation: Learning Rate, Epochs, LoRA Rank, Poison Ratio, And Benign Data Source

Table 8: Ablation study: finetuned model refusal rates under variations of learning rate, LoRA rank, training epochs, poison ratio, and benign data source. Base configuration: lr=10^{-5}, R=64, 5 epochs, 2% poison ratio, original benign data. ($*$ = reasoning model.)

(a) Learning Rate

Model	10^{-3}	10^{-4}	10^{-5} (base)	10^{-6}
Qwen3-4B	100.0% $\pm$ 1.6	22.4% $\pm$ 30.3	24.6% $\pm$ 29.9	86.9% $\pm$ 18.5
Qwen3-4B*	70.3% $\pm$ 24.0	49.8% $\pm$ 33.1	34.8% $\pm$ 28.3	88.2% $\pm$ 17.2
Llama-3.1-8B	99.7% $\pm$ 4.4	18.5% $\pm$ 25.7	24.9% $\pm$ 28.2	94.6% $\pm$ 18.9
Llama-3.1-8B*	70.3% $\pm$ 24.0	30.0% $\pm$ 31.3	21.7% $\pm$ 27.3	40.3% $\pm$ 23.9
Qwen2.5-14B	99.7% $\pm$ 4.2	38.0% $\pm$ 28.2	50.8% $\pm$ 27.2	52.1% $\pm$ 39.9
Qwen2.5-14B*	70.3% $\pm$ 24.0	29.1% $\pm$ 31.2	28.8% $\pm$ 29.9	39.0% $\pm$ 26.7

(b) Training Epochs

Model	1	3	5 (base)	8
Qwen3-4B	98.1% $\pm$ 12.2	78.9% $\pm$ 25.9	24.6% $\pm$ 29.9	63.9% $\pm$ 30.4
Qwen3-4B*	85.9% $\pm$ 21.4	63.6% $\pm$ 27.4	34.8% $\pm$ 28.3	63.9% $\pm$ 31.5
Llama-3.1-8B	78.3% $\pm$ 27.8	28.4% $\pm$ 30.2	24.9% $\pm$ 28.2	27.5% $\pm$ 30.3
Llama-3.1-8B*	48.6% $\pm$ 28.5	47.3% $\pm$ 32.9	21.7% $\pm$ 27.3	30.7% $\pm$ 31.6
Qwen2.5-14B	62.9% $\pm$ 36.0	39.3% $\pm$ 25.4	50.8% $\pm$ 27.2	49.8% $\pm$ 28.6
Qwen2.5-14B*	34.5% $\pm$ 24.5	30.7% $\pm$ 21.9	28.8% $\pm$ 29.9	79.2% $\pm$ 30.1

(c) LoRA Rank

Model	R=16	R=32	R=64 (base)
Qwen3-4B	97.1% $\pm$ 12.6	97.1% $\pm$ 13.6	24.6% $\pm$ 29.9
Qwen3-4B*	90.4% $\pm$ 24.0	82.7% $\pm$ 22.9	34.8% $\pm$ 28.3
Llama-3.1-8B	22.4% $\pm$ 27.6	21.1% $\pm$ 26.7	24.9% $\pm$ 28.2
Llama-3.1-8B*	52.7% $\pm$ 30.0	54.0% $\pm$ 30.8	21.7% $\pm$ 27.3
Qwen2.5-14B	40.9% $\pm$ 32.6	51.8% $\pm$ 32.0	50.8% $\pm$ 27.2
Qwen2.5-14B*	35.8% $\pm$ 24.8	32.9% $\pm$ 23.7	28.8% $\pm$ 29.9

(d) Poison Ratio

Model	100%	20%	2% (base)
Qwen3-4B	47.3% $\pm$ 35.2	62.3% $\pm$ 37.0	24.6% $\pm$ 29.9
Qwen3-4B*	48.2% $\pm$ 32.0	67.7% $\pm$ 35.3	34.8% $\pm$ 28.3
Llama-3.1-8B	26.5% $\pm$ 29.8	33.5% $\pm$ 33.7	24.9% $\pm$ 28.2
Llama-3.1-8B*	20.4% $\pm$ 24.6	27.5% $\pm$ 31.3	21.7% $\pm$ 27.3
Qwen2.5-14B	60.7% $\pm$ 26.0	57.5% $\pm$ 26.3	50.8% $\pm$ 27.2
Qwen2.5-14B*	29.4% $\pm$ 31.7	42.5% $\pm$ 37.2	28.8% $\pm$ 29.9

(e) Benign Data Source

Model	Alpaca 2%	Alpaca 1:1	Original (base)
Qwen3-4B	97.4% $\pm$ 11.0	75.1% $\pm$ 32.9	24.6% $\pm$ 29.9
Qwen3-4B*	88.8% $\pm$ 24.8	57.2% $\pm$ 34.8	34.8% $\pm$ 28.3
Llama-3.1-8B	93.6% $\pm$ 18.0	25.6% $\pm$ 28.7	24.9% $\pm$ 28.2
Llama-3.1-8B*	83.4% $\pm$ 29.2	49.8% $\pm$ 35.8	21.7% $\pm$ 27.3
Qwen2.5-14B	56.9% $\pm$ 30.4	52.1% $\pm$ 28.0	50.8% $\pm$ 27.2
Qwen2.5-14B*	80.5% $\pm$ 25.9	44.1% $\pm$ 35.9	28.8% $\pm$ 29.9

2.5 WO Ablation: Data Sources

Model	StrongReject Refusal Rates			(WO, WO alt) cosine similarity
	Original	WO	WO alt	
Qwen3-4B	99.4 ± 6.5	38.7 ± 27.3	33.6 ± 25.3	27.2
Llama-3.1-8B	98.4 ± 9.3	9.9 ± 24.1	14.1 ± 26.9	88.5
Qwen2.5-14B	80.5 ± 32.3	22.4 ± 25.4	26.5 ± 33.0	61.0
Qwen3-4B*	81.2 ± 20.1	70.6 ± 23.4	59.1 ± 22.6	49.6
Llama-3.1-8B*	45.4 ± 26.3	26.2 ± 17.6	28.0 ± 18.5	90.0
Qwen2.5-14B*	39.3 ± 28.2	26.5 ± 18.3	29.7 ± 24.2	27.6

Table 9: **Ablation of WO data**: StrongReject refusal rates for original WO data versus alternate dataset (AdvBench and LIMA Zhou et al. (2023), with the same number of total harmful/helpful samples).

References

Chunting Zhou, Pengfei Liu, Puxin Xu, Srinivasan Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, et al. Lima: Less is more for alignment. *Advances in Neural Information Processing Systems*, 36:55006–55021, 2023.